

Modeling Immune Response Gene Expression Using Two-Way ANOVA and Repeated Measures in SAS

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Abstract

This study analyzes immune response gene expression using Two-Way ANOVA and Repeated Measures models implemented in SAS. The dataset (GSE71165) consists of RNA-seq gene expression measurements under control (CTRL) and immune (IMM) conditions with technical replicates. The objective is to evaluate the effects of experimental condition and replicate on gene expression while accounting for within-gene correlation.

Exploratory data analysis was conducted to assess data structure, distribution, and variability. Two-Way ANOVA was applied to test main and interaction effects assuming independence, while a repeated-measures mixed model was used to account for correlation among repeated observations within genes.

Results indicate that Two-Way ANOVA fails to detect a significant condition effect, whereas the repeated-measures model reveals a statistically significant but small condition effect. Strong replicate effects and significant interaction terms suggest the presence of substantial technical variability, highlighting the importance of using appropriate statistical models for correlated gene expression data.

Keywords: Two-Way ANOVA, Repeated Measures, Gene Expression, SAS, Statistical Modeling, Interaction Effects

1 Introduction

Gene expression analysis is essential in understanding biological responses to treatments, environmental conditions, and time-dependent processes. Statistical modeling allows researchers to determine whether observed variations in gene expression are statistically significant.

Two-Way ANOVA is commonly used to evaluate the effects of two categorical independent variables and their interaction on a continuous response variable. When measurements are collected repeatedly from the same subjects, Repeated Measures ANOVA is applied to account for correlation within subjects.

This study applies both approaches to model immune response gene expression data using SAS.

2 Data Description and Exploratory Analysis

The dataset used in this study was obtained from the Gene Expression Omnibus (GEO) database under accession number GSE71165 [1]. This dataset contains RNA sequencing (RNA-seq) measurements of gene expression under control (CTRL) and immune-stimulated (IMM) conditions, with two technical replicates per condition.

The dataset consists of 30,043 genes and 7 variables, including gene identifiers, expression measurements (FPKM values), and a q-value representing differential expression significance between conditions. RNA-seq data are known to exhibit high variability and skewness, motivating the use of log-transformed expression values.

Exploratory data analysis (EDA) was conducted using SAS to examine dataset structure, missing values, summary statistics, and distributional properties. Log transformation was applied to stabilize variance and improve normality assumptions.

2.1 Exploratory Analysis Workflow

The following SAS procedures were used for EDA:

- Data import and structure inspection (PROC IMPORT, PROC CONTENTS)
- Missing value assessment (PROC MEANS)
- Summary statistics (PROC MEANS)
- Dataset dimensions (PROC SQL)
- Visualization (boxplots and correlation analysis)

3 Methods

3.1 Two-Way ANOVA model

Two-Way ANOVA was used to evaluate the main effects of two categorical experimental factors and their interaction on gene expression. Let Y_{ijk} denote the gene expression response for the k th observation in factor level i of factor A and level j of factor B. The model may be written as

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}, \quad (1)$$

where μ is the overall mean, α_i is the effect of factor A, β_j is the effect of factor B, $(\alpha\beta)_{ij}$ is the interaction effect, and ε_{ijk} is the random error term.

The SAS procedure PROC GLM was used to estimate the model and test the significance of main and interaction effects.

3.2 Repeated Measures Mixed Model

Because multiple observations were collected from the same gene, a repeated-measures mixed-effects model was used to account for within-gene correlation. Let Y_{ijk} denote the log-transformed expression for gene k under condition i and replicate j . The model is given by:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + u_k + \varepsilon_{ijk}, \quad (2)$$

where μ is the overall mean, α_i is the effect of condition, β_j is the effect of replicate, $(\alpha\beta)_{ij}$ is the interaction effect, u_k is a random effect for gene k , and ε_{ijk} represents within-gene residual error.

A compound symmetry covariance structure was assumed to model correlation among repeated observations within each gene. The model was fitted using the SAS procedure PROC MIXED.

3.3 SAS modeling workflow

Algorithm 1 SAS workflow for immune response gene expression analysis

- 1: Import the gene expression dataset into SAS
 - 2: Inspect variable structure and summary statistics
 - 3: Produce exploratory plots and descriptive summaries
 - 4: Fit a Two-Way ANOVA model with main effects and interaction
 - 5: Evaluate significance of factor A, factor B, and interaction terms
 - 6: Fit a Repeated Measures ANOVA model for within-subject observations
 - 7: Check model assumptions and diagnostic output
 - 8: Compare inferential conclusions from both modeling strategies
 - 9: Summarize results in tables and figures
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4 Results

4.1 Two-Way ANOVA findings

The Two-Way ANOVA results indicate that the effect of condition (CTRL vs IMM) on log-transformed gene expression is not statistically significant ($p = 0.6347$), suggesting no overall difference in mean expression between conditions. However, the effect of replicate is statistically significant ($p = 0.0190$), indicating the presence of technical variability between replicate measurements. The interaction between condition

and replicate is not significant ($p = 0.1592$), implying that the effect of condition is consistent across replicates.

The extremely low R^2 value suggests that the model explains negligible variation in gene expression. This result is expected because gene expression responses are gene-specific, and aggregating all genes into a single analysis may obscure meaningful biological differences.

4.2 Repeated Measures findings

The repeated-measures mixed model provides a more appropriate analysis by accounting for within-gene correlation. The effect of condition is statistically significant ($p = 0.0241$), indicating a difference in gene expression between control and immune conditions when within-subject dependence is considered. However, the estimated effect size is small.

The effect of replicate is highly significant ($p < 0.0001$), demonstrating strong technical variability between measurements. Additionally, the interaction between condition and replicate is statistically significant ($p < 0.0001$), indicating that the effect of condition is not consistent across replicates.

These results suggest that although a biological signal exists, it is relatively weak compared to technical variability, and the presence of interaction effects raises concerns about the stability of measurements across replicates.

4.3 Figure for model results

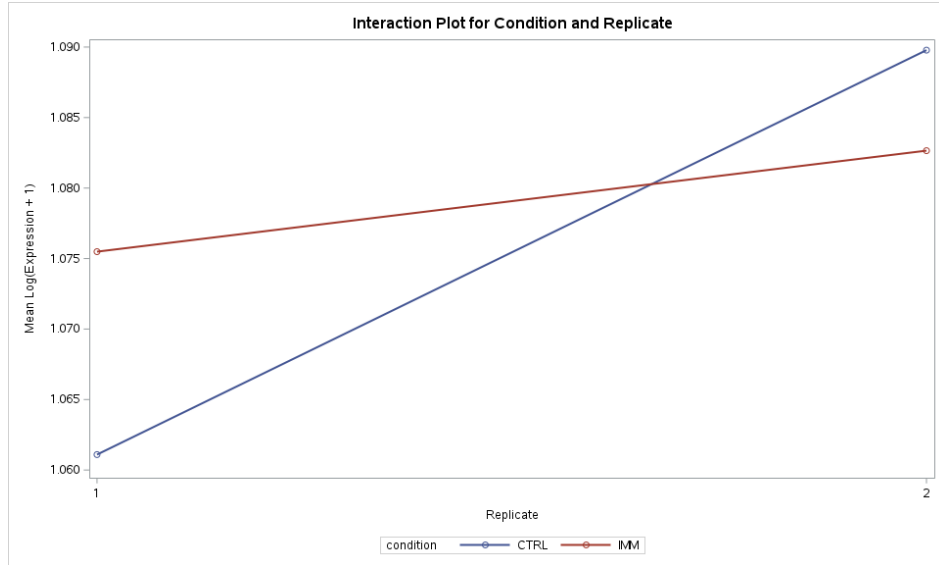


Fig. 1 Interaction plot of mean log-transformed gene expression by condition and replicate.

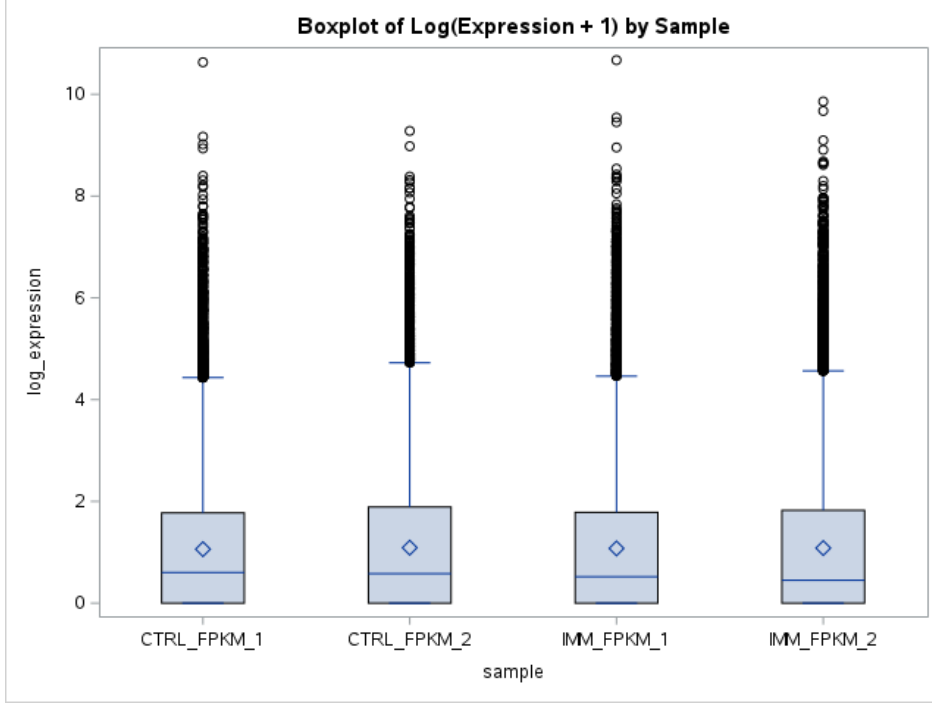


Fig. 2 Boxplot of log-transformed gene expression across conditions and replicates.

5 Discussion

This study compares Two-Way ANOVA and repeated-measures mixed modeling approaches for analyzing RNA-seq gene expression data. The results highlight important differences in statistical inference when correlation among observations is properly accounted for.

The Two-Way ANOVA model assumes independence among observations and fails to detect a significant condition effect. In contrast, the repeated-measures model accounts for within-gene correlation and reveals a statistically significant condition effect, demonstrating the importance of appropriate model selection.

However, the magnitude of the condition effect is small, and strong replicate effects indicate substantial technical variability. The presence of a significant interaction between condition and replicate suggests that measurements are not fully consistent across replicates, which may affect the reliability of biological conclusions.

A key limitation of this analysis is that gene expression responses are inherently gene-specific. Aggregating all genes into a single model may obscure differential expression patterns. More advanced approaches, such as gene-level differential expression analysis, may provide more biologically meaningful insights.

Overall, the findings emphasize the importance of accounting for repeated measurements and technical variability when analyzing high-dimensional biological data.

6 Conclusion

This project applies SAS-based Two-Way ANOVA and Repeated Measures ANOVA to the study of immune response gene expression. The analysis is designed to investigate main effects, interaction effects, and within-subject variation in a structured experimental setting.

By comparing these complementary statistical approaches, the study aims to provide a clearer understanding of the factors that influence gene expression patterns in immune response data. This project applies SAS-based Two-Way ANOVA and a repeated-measures mixed model to immune response gene expression data. The analysis demonstrates that accounting for within-gene correlation can substantially alter statistical conclusions. While the Two-Way ANOVA did not detect a significant condition effect, the repeated-measures model identified a statistically significant but small effect after accounting for dependence among observations. Strong replicate and interaction effects indicate substantial technical variability, emphasizing the importance of appropriate modeling in high-dimensional biological data.

Supplementary information. Supplementary material may include SAS code, ANOVA output tables, repeated measures diagnostics, and additional figures not included in the main text.

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Declarations

Funding

No external funding was reported for this study.

Conflict of interest/Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

The gene expression dataset analyzed in this study is publicly available from the Gene Expression Omnibus under accession GSE71165. SAS code and derived outputs may be provided by the authors upon request.

Materials availability

Not applicable.

Code availability

The SAS code used for preprocessing, modeling, and analysis is available from the authors and may be included in supplementary materials.

Author contribution

William Lorenzo contributed to statistical analysis, interpretation of results, and manuscript drafting. Syed Muhammad Arham Ali contributed to project development,

review, and interpretation. Albert Hickey contributed to project development, review, and interpretation.

Appendix A Reproducibility Notes

Key implementation details from the SAS workflow are summarized below:

- Software: SAS
- Main procedures: PROC IMPORT, PROC CONTENTS, PROC MEANS, PROC SQL, PROC PRINT, PROC UNIVARIATE, PROC GLM, PROC MIXED
- Analysis types: Two-Way ANOVA and repeated-measures mixed modeling
- Response variable: $\log(\text{expression} + 1)$
- Main objective: test factor effects, interaction effects, and within-subject variation

Appendix B Figure File Names

To maintain clarity and consistency, only the most relevant figures used in the final analysis are listed below:

- `log_expression_boxplot.png`
- `anova_interaction_plot.png`

References

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